

Error Considerations

Absolute and Relative Errors

Therefore, absolute error

$$E_a = |X_E - X_A|$$

and relative error is defined by

$$E_r = \left| \frac{X_E - X_A}{X_E} \right|$$

provided $X_E \neq 0$ or X_E is not too close to zero. The percentage relative error is

$$E_p = 100E_r.$$

If X_E is the exact or true value and X_A is an approximation to X_E , then X_A is said to approximate X_E to t significant figures if t is the largest nonnegative integer for which

$$\frac{X_E - X_A}{|X_E|} < 5 \times 10^{-t}$$

Truncation Errors

Taylor Series

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)(x-a)^k}{k!}$$

Truncated After n Terms

$$f_n(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(a)(x-a)^k}{k!}$$

Truncation Error (Tail Term)

$$E_n(x) = f(x) - f_n(x) \approx \frac{f^{(n)}(\beta)(x-a)^n}{n!},$$

for some β between a and x .

Error Bound

If

$$M_n(x) = \max_{a \leq \beta \leq x} |f^{(n)}(\beta)|,$$

then

$$|E_n(x)| \leq \frac{M_n(x)|x-a|^n}{n!}.$$

1. Root-Finding Methods

Let $f(x)$ be continuous. Goal: solve $f(x) = 0$.

1.1 Key Formulas

Bisection (bracketing): if $f(a)f(b) < 0$, then for $n = 0, 1, \dots$:

$$c_n = \frac{a_n + b_n}{2}, \quad \text{choose } [a_{n+1}, b_{n+1}]$$

where $f(a_{n+1})f(b_{n+1}) < 0$.

Error bound after n steps:

$$|c_n - \alpha| \leq \frac{b_0 - a_0}{2^{n+1}}.$$

False Position (Regula Falsi):

$$x_{\text{new}} = b - f(b) \frac{b - a}{f(b) - f(a)}.$$

Newton–Raphson (open method):

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Secant (no derivative, uses two points):

$$x_{n+1} = x_n - f(x_n) \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}.$$

1.2 Example (Bisection + Newton)

Question. Solve $f(x) = x^3 - x - 2 = 0$:

- (a) Use the bisection method for 4 iterations starting with $[1, 2]$.
- (b) Perform 2 Newton iterations starting from $x_0 = 1.5$.

Solution.

(a) Bisection on $[1, 2]$.

Compute

$$f(1) = 1 - 1 - 2 = -2 < 0, \quad f(2) = 8 - 2 - 2 = 4 > 0,$$

so a root lies in $[1, 2]$.

Iteration n :

$$c_n = \frac{a_n + b_n}{2}, \quad \text{check sign of } f(c_n).$$

- **Iteration 1:** $c_1 = \frac{1+2}{2} = 1.5$.

$$f(1.5) = 3.375 - 2.25 - 2 = -0.875 < 0.$$

So root in $[1.5, 2]$. Set $a_2 = 1.5, b_2 = 2$.

- **Iteration 2:** $c_2 = \frac{1.5+2}{2} = 1.75$.

$$f(1.75) = 5.359375 - 3.0625 - 2 = 0.296875 > 0.$$

So root in $[1.5, 1.75]$. Set $a_3 = 1.5, b_3 = 1.75$.

- **Iteration 3:** $c_3 = \frac{1.5+1.75}{2} = 1.625$.

$$f(1.625) = 4.2910 - 2.6406 - 2 \approx -0.3496 < 0.$$

So root in $[1.625, 1.75]$.

- **Iteration 4:** $c_4 = \frac{1.625+1.75}{2} = 1.6875$.

$$f(1.6875) \approx 4.805 - 2.847 - 2 = -0.042 \text{ (approx.)}.$$

So root in $[1.6875, 1.75]$.

After 4 iterations, approximate root $x \approx c_4 = 1.6875$ (true root ≈ 1.5214 ; more iterations are needed for high accuracy).

(b) Newton–Raphson from $x_0 = 1.5$.

Here

$$f(x) = x^3 - x - 2, \quad f'(x) = 3x^2 - 1.$$

Iteration 1:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 1.5 - \frac{-0.875}{3(1.5)^2 - 1} = 1.6522.$$

Iteration 2: evaluate $f(1.6522)$ and $f'(1.6522)$:

$$f(1.6522) \approx 1.6522^3 - 1.6522 - 2 \approx 0.852,$$

$$f'(1.6522) = 3(1.6522)^2 - 1 \approx 3(2.73) - 1 \approx 7.19.$$

Then

$$x_2 = 1.6522 - \frac{0.852}{7.19} \approx 1.6522 - 0.1186 = 1.5336.$$

Continuing a few more iterations gives $x \approx 1.5214$ very quickly.

2. Linear Systems $A\mathbf{x} = \mathbf{b}$

2.1 Key Methods

Gaussian elimination (naive):

- Forward elimination: use row operations to transform $[A|b] \rightarrow [U|d]$ where U is upper triangular.
- Back substitution: solve $U\mathbf{x} = \mathbf{d}$ from last row upwards.

$$\begin{bmatrix} a_{(11)}^{(1)} & a_{(12)}^{(1)} & \cdots & a_{(1n)}^{(1)} \\ a_{(21)}^{(1)} & a_{(22)}^{(1)} & \cdots & a_{(2n)}^{(1)} \\ \vdots & \vdots & \vdots & \vdots \\ a_{(n1)}^{(1)} & a_{(n2)}^{(1)} & \cdots & a_{(nn)}^{(1)} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1^{(1)} \\ b_2^{(1)} \\ \vdots \\ b_n^{(1)} \end{bmatrix}$$

LU factorization:

- Factor $A = LU$, where L is lower triangular, U upper triangular.
- Solve $Ly = b$ (forward) then $Ux = y$ (backward).

A can be factored into the product of a lower triangular matrix L and an upper triangular matrix U , i.e., $A = LU$, where

$$\begin{bmatrix} l_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ l_{31} & l_{32} & l_{33} & 0 \\ l_{41} & l_{42} & l_{43} & l_{44} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} & u_{14} \\ 0 & 1 & u_{23} & u_{24} \\ 0 & 0 & 1 & u_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

2.2 Example (Gaussian Elimination)

Question Use Gaussian elimination and LU method to solve the system

$$\begin{cases} x_1 + 2x_2 + x_3 = 0, \\ 2x_1 + 2x_2 + 3x_3 = 3, \\ -x_1 - 3x_2 + 0x_3 = 2. \end{cases}$$

Solution.

Augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 2 & 2 & 3 & 3 \\ -1 & -3 & 0 & 2 \end{array} \right].$$

Step 2: Eliminate below the first pivot

$$R_2 \leftarrow R_2 - 2R_1 \quad \Rightarrow \quad [0 \quad -2 \quad 1 \mid 3]$$

$$R_3 \leftarrow R_3 + R_1 \quad \Rightarrow \quad [0 \quad -1 \quad 1 \mid 2]$$

New matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -2 & 1 & 3 \\ 0 & -1 & 1 & 2 \end{array} \right]$$

3. Interpolation

$$a_0 + a_1x_0 + a_2x_0^2 + \dots + a_nx_0^n = y_0$$

$$a_0 + a_1x_1 + a_2x_1^2 + \dots + a_nx_1^n = y_1$$

$$\vdots = \vdots$$

$$a_0 + a_1x_n + a_2x_n^2 + \dots + a_nx_n^n = y_n$$

$$Xa = Y$$

$$X = \begin{bmatrix} 1 & x_0 & \cdots & x_0^{n-1} & x_0^n \\ 1 & x_1 & \cdots & x_1^{n-1} & x_1^n \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_n & \cdots & x_n^{n-1} & x_n^n \end{bmatrix}, \quad a = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix}, \quad Y = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_n \end{bmatrix}$$

3.1 Lagrange Polynomial

Given $(x_0, y_0), \dots, (x_n, y_n)$ with distinct x_i , the interpolating polynomial is

$$p_n(x) = \sum_{i=0}^n y_i L_i(x), \quad L_i(x) = \prod_{\substack{0 \leq j \leq n \\ j \neq i}} \frac{x - x_j}{x_i - x_j}.$$

Example For $x_0 = 1, x_1 = 2, x_2 = 3, x_3 = 4$, the following polynomials satisfy the above property:

$$L_0(x) = \frac{(x-2)(x-3)(x-4)}{(1-2)(1-3)(1-4)},$$

$$L_1(x) = \frac{(x-1)(x-3)(x-4)}{(2-1)(2-3)(2-4)},$$

$$L_2(x) = \frac{(x-1)(x-2)(x-4)}{(3-1)(3-2)(3-4)},$$

$$L_3(x) = \frac{(x-1)(x-2)(x-3)}{(4-1)(4-2)(4-3)}.$$

3.2 Newton Divided Differences

x	x_0	x_1	$\dots$	x_n
$f(x)$	$f(x_0)$	$f(x_1)$	$\dots$	$f(x_n)$

$$f[x_0] = f(x_0)$$

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0}$$

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$$

$$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0}$$

x_i	$f[x_i]$	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$
x_0	$f[x_0]$		
		$f[x_0, x_1]$	
x_1	$f[x_1]$		$f[x_0, x_1, x_2]$
		$f[x_1, x_2]$	
x_2	$f[x_2]$		$f[x_1, x_2, x_3]$
		$f[x_2, x_3]$	
x_3	$f[x_3]$		$f[x_2, x_3, x_4]$
		$f[x_3, x_4]$	
x_4	$f[x_4]$		$f[x_3, x_4, x_5]$
		$f[x_4, x_5]$	
x_5	$f[x_5]$		
$\vdots$	$\vdots$	$\vdots$	$\vdots$

Newton form:

$$p_n(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + \cdots .$$

3.3 Example

Question Find a polynomial of order 2 that interpolates the table

x	0.2	0.4	0.6
y	-0.95	-0.82	-0.65

Solution.
We look for a quadratic polynomial

$$p(x) = ax^2 + bx + c$$

that passes through the three points

$$(0.2, -0.95), \quad (0.4, -0.82), \quad (0.6, -0.65).$$

Substituting each point into $p(x)$ gives a system for a , b , c :

$$x = 0.2 : \quad a(0.2)^2 + b(0.2) + c = -0.95 \tag{1}$$

$$x = 0.4 : \quad a(0.4)^2 + b(0.4) + c = -0.82 \tag{2}$$

$$x = 0.6 : \quad a(0.6)^2 + b(0.6) + c = -0.65 \tag{3}$$

Subtract (1) from (2), and (2) from (3) to eliminate c :

$$(2) - (1) : \quad 0.12a + 0.2b = 0.13 \tag{4}$$

$$(3) - (2) : \quad 0.20a + 0.2b = 0.17 \tag{5}$$

Now subtract (4) from (5):

$$0.08a = 0.04 \quad \Rightarrow \quad a = \frac{0.04}{0.08} = 0.5.$$

Substitute $a = 0.5$ into (5):

$$0.20(0.5) + 0.2b = 0.17 \quad \Rightarrow \quad 0.10 + 0.2b = 0.17 \quad \Rightarrow \quad b = 0.35.$$

Use equation (1) to find c :

$$0.04(0.5) + 0.2(0.35) + c = -0.95 \quad \Rightarrow \quad c = -1.04.$$

Hence the interpolating quadratic polynomial is

$$p(x) = 0.5x^2 + 0.35x - 1.04$$

Example Find the polynomial that interpolates the table

x	4.5	6.1
y	7.1	2.3

We are given two data points

$$(4.5, \; 7.1), \qquad (6.1, \; 2.3).$$

Since there are only two points, the interpolating polynomial is linear. The slope is

$$m = \frac{2.3 - 7.1}{6.1 - 4.5} = \frac{-4.8}{1.6} = -3.$$

Thus the interpolating polynomial is

$$p(x) = 7.1 - 3(x - 4.5). \\ p(x) = 20.6 - 3x.$$

4. Numerical Integration

Let $I = \int_a^b f(x) \, dx$, $h = \frac{b-a}{n}$, $x_i = a + ih$.

4.1 Trapezoidal Rule

Single interval:

$$I \approx \frac{h}{2} [f(a) + f(b)].$$

Composite (n subintervals):

$$\int_a^b f(x) \, dx = \frac{h}{2} [f(a) + f(b)] - \frac{1}{12}(h)^3 f''(\xi).$$

4.2 Simpson's 1/3 Rule

Requires even n . Composite formula:

$$I_{S_{1/3}} = \int_a^b f(x) \, dx \approx \frac{h}{2} (f_1 + 4f_2 + 2f_3 + \cdots + 4f_n + f_{n+1}) \\ - \frac{1}{90} \left(\frac{b-a}{2} \right)^5 f^{(4)}(\xi)$$

4.3 Simpson's 3/8 Rule

For $n = 3$ subintervals (single panel):

$$I_{S_{3/8}} \approx \frac{3h}{8} (f_1 + 3f_2 + 3f_3 + f_4) + \frac{3h^5}{80} f^{(4)}(\xi)$$

4.4 Example (Integrate x^2 on $[0, 1]$)

Question. Approximate $\int_0^1 x^2 \, dx$ using:

- (a) composite trapezoidal rule with $n = 4$,
- (b) composite Simpson's 1/3 rule with $n = 4$.

Solution.

Here $f(x) = x^2$, $a = 0$, $b = 1$, $n = 4$, so

$$h = \frac{1-0}{4} = 0.25, \quad x_i = 0, 0.25, 0.5, 0.75, 1.$$

Function values:

$$f(0) = 0, \; f(0.25) = 0.0625, \; f(0.5) = 0.25,$$

$$f(0.75) = 0.5625, \; f(1) = 1.$$

(a) Trapezoidal (composite).

$$I_T \approx \frac{h}{2} [f(0) + 2(f(0.25) + f(0.5) + f(0.75)) + f(1)].$$

Sum:

$$f(0.25) + f(0.5) + f(0.75) = 0.875.$$

Thus

$$I_T \approx \frac{0.25}{2} [0 + 2(0.875) + 1] = 0.125(1.75 + 1) = 0.34375.$$

(b) Simpson's 1/3 (composite).

$$I_S \approx \frac{h}{3} [f(0) + f(1) + 4(f(0.25) + f(0.75)) + 2f(0.5)].$$

Compute:

$$f(0.25) + f(0.75) = 0.625.$$

Then

$$I_S \approx \frac{0.25}{3} [0 + 1 + 4(0.625) + 2(0.25)] = \frac{1}{3} \approx 0.33333.$$

Exact integral:

$$\int_0^1 x^2 \, dx = \left. \frac{x^3}{3} \right|_0^1 = \frac{1}{3}.$$

So Simpson's rule is exact here.

5. Numerical Differentiation

Use step size h .

5.1 First Derivative Formulas

Forward difference:

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}.$$

Backward difference:

$$f'(x) \approx \frac{f(x) - f(x-h)}{h}.$$

Central difference:

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}.$$

5.2 Example ($f(x) = e^x$ at $x = 0$)

Question. For $f(x) = e^x$, approximate $f'(0)$ using forward, backward, and central differences with $h = 0.1$. Compare with exact value.

Solution.

Exact derivative: $f'(x) = e^x$, so $f'(0) = 1$.

Compute function values:

$$f(0) = e^0, \quad f(0.1) = e^{0.1}, \quad f(-0.1) = e^{-0.1}.$$

Forward difference:

$$f'(0) \approx \frac{f(0.1) - f(0)}{0.1} = \frac{1.10517 - 1}{0.1} \approx 1.0517.$$

Backward difference:

$$f'(0) \approx \frac{f(0) - f(-0.1)}{0.1} = \frac{1 - 0.90484}{0.1} \approx 0.9516.$$

Central difference:

$$f'(0) \approx \frac{f(0.1) - f(-0.1)}{2(0.1)} = \frac{1.10517 - 0.90484}{0.2} \approx 1.0017.$$

Central difference is much closer to the exact value 1.

6. ODE Initial-Value Problems (IVPs)

Consider $y'(t) = f(t, y)$, $y(t_0) = y_0$, step size

$$h = \frac{t_{n+k} - t_n}{k}$$

6.1 Euler's Method

$$y_{n+1} = y_n + h f(t_n, y_n).$$

First-order accurate.

6.2 Runge–Kutta 2 (Midpoint)

$$k_1 = f(t_n, y_n),$$

$$k_2 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right),$$

$$y_{n+1} = y_n + h k_2.$$

6.3 Runge–Kutta 4 (RK4)

$$k_1 = f(t_n, y_n),$$

$$k_2 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right),$$

$$k_3 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right),$$

$$k_4 = f\left(t_n + h, y_n + h k_3\right),$$

$$y_{n+1} = y_n + \frac{h}{6} (k_1 + 2k_2 + 2k_3 + k_4).$$

6.4 Example (Euler & RK2)

Question. Solve the IVP

$$y' = 2t - y, \quad y(0) = 1$$

approximately at $t = 0.4$ using:

- (a) Euler's method with $h = 0.2$,
- (b) RK2 (midpoint) with $h = 0.2$.

Solution. We take two steps: $t_0 = 0$, $t_1 = 0.2$, $t_2 = 0.4$.

(a) Euler.
Step 1 ($t_0 = 0$, $y_0 = 1$):

$$f(t_0, y_0) = 2 \cdot 0 - 1 = -1,$$

$$y_1 = y_0 + h f(t_0, y_0) = 1 + 0.2(-1) = 0.8.$$

Step 2 ($t_1 = 0.2$, $y_1 = 0.8$):

$$f(t_1, y_1) = 2(0.2) - 0.8 = -0.4,$$

$$y_2 = y_1 + h f(t_1, y_1) = 0.8 + 0.2(-0.4) = 0.72.$$

Thus Euler gives $y(0.4) \approx 0.72$.

(b) RK2 (midpoint).
Step 1 ($t_0 = 0$, $y_0 = 1$):

$$k_1 = f(0, 1) = -1,$$

midpoint:

$$t_{0,\text{mid}} = 0 + \frac{h}{2} = 0.1, \quad y_{0,\text{mid}} = 1 + \frac{h}{2} k_1 = 1 + 0.1(-1) = 0.9,$$

$$k_2 = f(0.1, 0.9) = 2(0.1) - 0.9 = 0.2 - 0.9 = -0.7,$$

$$y_1 = y_0 + h k_2 = 1 + 0.2(-0.7) = 1 - 0.14 = 0.86.$$

Step 2 ($t_1 = 0.2$, $y_1 = 0.86$):

$$k_1 = f(0.2, 0.86) = 0.4 - 0.86 = -0.46,$$

$$t_{1,\text{mid}} = 0.3, \quad y_{1,\text{mid}} = 0.86 + 0.1(-0.46) = 0.86 - 0.046 = 0.814,$$

$$k_2 = f(0.3, 0.814) = 0.6 - 0.814 = -0.214,$$

$$y_2 = y_1 + h k_2 = 0.86 + 0.2(-0.214) = 0.86 - 0.0428 = 0.8172.$$

Thus RK2 gives $y(0.4) \approx 0.8172$, significantly more accurate than Euler (true solution ≈ 0.819).

More Example

Solve the following system of equations by Crout's Method:

$$\begin{cases} 2x - 4y + z = 4, \\ 6x + 2y - z = 10, \\ -2x + 6y - 2z = -6. \end{cases}$$

In matrix form,

$$\begin{bmatrix} 2 & -4 & 1 \\ 6 & 2 & -1 \\ -2 & 6 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 10 \\ -6 \end{bmatrix}.$$

Step 1: Write $A = LU$ with

$$A = \begin{bmatrix} 2 & -4 & 1 \\ 6 & 2 & -1 \\ -2 & 6 & -2 \end{bmatrix}, \quad L = \begin{bmatrix} \ell_{11} & 0 & 0 \\ \ell_{21} & \ell_{22} & 0 \\ \ell_{31} & \ell_{32} & \ell_{33} \end{bmatrix}, \quad U = \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}.$$

Step 2: Multiply out LU and equate to A :

$$\begin{bmatrix} 2 & -4 & 1 \\ 6 & 2 & -1 \\ -2 & 6 & -2 \end{bmatrix} = \begin{bmatrix} \ell_{11} u_{12} & \ell_{11} u_{13} \\ \ell_{21} u_{12} + \ell_{22} u_{23} & \ell_{21} u_{13} + \ell_{22} u_{23} \\ \ell_{31} u_{12} + \ell_{32} u_{23} & \ell_{31} u_{13} + \ell_{32} u_{23} + \ell_{33} \end{bmatrix}.$$